BUY-SIDE EQUITY RESEARCH — SCREEN

What Hedges Semiconductors

Sector-first screen for negative correlation to SOXX, measured through the drawdown

SCREEN | 7 sectors, 13 names | window includes the July 2026 chip selloff

1 EXECUTIVE SUMMARY

This screen inverts the previous one. Rather than asking what is uncorrelated to the broad market, it asks the operational question: what actually rises when semiconductors fall? The work is built top-down as requested — sector level first, then the patterns that explain the sector results, then individual names — and critically, the measurement window now runs through the drawdown itself (14 May to 27 July 2026), so this is an out-of-sample test rather than a theory.

The regime diagnosis comes first because it determines whether any of this works. The current drawdown is semiconductor-specific, not systemic: SOXX has fallen about 19.5% from its 22 June peak while the broad market has held up and defensive sectors have actively risen. That is a rotation reversal, and it is the regime in which negatively-correlated exposure pays. It is not a liquidity crash, where correlations would converge toward 1 and nothing here would protect you.

The central finding is that the conventional 'defensive' label is now unreliable as a guide to chip hedging, because AI capital expenditure has annexed two of the traditional defensive sectors. Utilities, the classic bond-proxy defensive, is now a power-for-data-centres trade and its correlation to SOXX has collapsed to just -0.15. Industrials is outright POSITIVE at +0.62 — it has become an AI-infrastructure proxy. The sectors that actually hedge chips are the ones with no AI capex linkage at all: consumer staples (-0.63), health care (-0.39), energy (-0.29) and financials (-0.26).

Window: 14 May – 27 Jul 2026	Observations: 48 daily returns (24 in drawdown)
SOXX drawdown: -19.5% from 22 Jun peak	SOXX volatility: 67.0% annualised
Best sector hedge: XLP staples, corr -0.63	In drawdown only: XLP -0.77, XLV -0.56
Best risk-adjusted: XLV health care, +11.9%	Failed 'defensives': XLI +0.62, XLU only -0.15
Top names: PGR -0.66, MCK -0.65, PM -0.38	Regime: Chip-specific unwind, not systemic

The practical read for a chip-heavy book: staples and health care are the reliable sector-level hedges, and within them Progressive, McKesson, Philip Morris and Gilead are the measured names. But note the volatility asymmetry in Section 7 — SOXX runs at 67% annualised volatility against roughly 18% for these sectors, so a dollar of staples does not offset a dollar of chips. Negative correlation tells you the direction; it does not size the hedge.

2 REGIME DIAGNOSIS — WHICH KIND OF DRAWDOWN IS THIS

ELI5 — Two very different kinds of falling market

Imagine a crowded restaurant where everyone suddenly decides the fish is overpriced. They stop ordering fish and order steak instead — the fish stall empties, the steak stall booms. That is a ROTATION: money leaves one thing and goes into another, so the other thing goes UP. Now imagine instead that a fire alarm goes off and everyone runs for the exit, abandoning fish and steak alike. That is a CRASH: everything is sold at once and nothing protects you. Hedges built on rotation logic work brilliantly in the first case and fail in the second. Diagnosing which one you are in is the whole game.

The evidence says this is a rotation, and a violent one. The semiconductor complex has given back roughly a fifth of its value from the 22 June peak, with the catalysts specific to chips rather than to the economy: a Broadcom guidance disappointment, reports of SK Hynix slowing high-bandwidth-memory (HBM) expansion, Intel's 18A foundry yield timeline slipping, Meta announcing it would build and sell its own compute, and a record one-day fall in SK Hynix that cascaded through Asia. Valuation and positioning unwound after a first half in which the group rose more than 80%.

Meanwhile the rest of the market did not follow it down. The Dow reached record intraday levels around the start of the sell-off, breadth broadened, the equal-weighted S&P outperformed the cap-weighted index, and defensive sectors led. Our own measurement confirms it directly: over the drawdown sub-period from the 22 June peak, staples rose 3.9%, utilities 2.1%, financials 5.9%, energy 8.0% and health care 8.9% while chips fell about 19.5%. Money left semiconductors and went somewhere — this screen identifies where.

The forward-looking caveat is the important one. Everything measured here describes capital rotating between equity sectors. If the chip unwind broadens into a genuine growth scare or a funding event — if the AI capex cycle is judged to be ending rather than pausing, and the earnings of the industrial and power complex are re-rated with it — the correlations in this report will move toward positive together. Section 8 treats this as the primary risk rather than a footnote.

3 METHODOLOGY

ELI5 — Reading a correlation to SOXX

Think of SOXX — the iShares Semiconductor ETF, the chip complex in one ticker — as one dancer, and each sector as another. A correlation of +1.0 means they step in perfect unison; 0.0 means they ignore each other; -1.0 means every time one steps left the other steps right. For a book that is already long chips, you want partners who reliably step the other way, because those are the positions that rise on the days your main book is being marked down.

We computed Pearson correlations of daily percentage returns between SOXX and each sector, using split- and dividend-adjusted closing prices pulled ticker by ticker and recomputed in pandas. Every series was spot-checked by recomputing daily returns against the vendor's published change figures; all eight checks matched exactly. Sector exposure is taken through the State Street Select Sector SPDR exchange-traded funds (ETFs) — XLP staples, XLV health care, XLE energy, XLF financials, XLU utilities, XLI industrials, XLY consumer discretionary — because they are the cleanest liquid proxy for sector behaviour and are what most desks trade for sector tilts.

Two windows are reported throughout. The FULL window (14 May to 27 July, 48 overlapping daily observations) captures both the final leg of the chip melt-up and the unwind. The DRAWDOWN sub-period (from the 22 June SOXX peak, 24 observations) isolates the conditions that actually matter for a hedge. Where the two disagree, the drawdown figure is the more decision-relevant one and the divergence is itself informative.

Statistical honesty: with 48 observations the standard error on a correlation is roughly 0.14, and with the 24-observation drawdown sub-period it is roughly 0.20. Differences of 0.1 between sectors are noise; the gap between staples at -0.63 and industrials at +0.62 is not. Beta figures are the regression slope of sector returns on SOXX returns and are structurally small here because SOXX volatility is roughly three to four times sector volatility — this is a sizing fact, not a weak signal, and Section 7 deals with it.

4 SECTOR LAYER — THE MEASURED RESULT

The table below ranks all seven measured sectors by correlation to SOXX over the full window, most negative first. 'Drawdown corr' and 'Drawdown ret' cover the sub-period from the 22 June chip peak — the conditions that matter. Beta is to SOXX. Volatility is annualised. Full-window returns for SOXX itself were roughly flat (-0.2%) because the melt-up and the unwind cancelled; the drawdown column is where the signal lives.

Sector	Corr SOXX	Drawdown corr	Drawdown ret %	Beta	Vol %	Window ret %
XLP Staples	-0.63	-0.77	+3.9	-0.16	17.8	+1.1
XLV Health care	-0.39	-0.56	+8.9	-0.11	18.8	+11.9
XLE Energy	-0.29	-0.23	+8.0	-0.10	23.6	+1.2
XLF Financials	-0.26	-0.35	+5.9	-0.05	13.9	+11.3
XLU Utilities	-0.15	-0.33	+2.1	-0.04	17.2	+2.4
XLY Discretionary	+0.34	+0.08	-3.6	+0.11	22.6	-6.4
XLI Industrials	+0.62	+0.64	+0.8	+0.17	18.6	+5.2
SOXX Semis	1.00	1.00	-19.5	1.00	67.0	-0.2

Consumer staples is the cleanest sector hedge in the book. Its correlation to chips is -0.63 across the whole window and strengthens to -0.77 during the drawdown itself — the single most negative reading anywhere in this study — while delivering a positive 3.9% return over the period chips fell a fifth. This is the behaviour a hedge is supposed to exhibit: more negatively correlated precisely when it is needed.

Health care is the best risk-adjusted answer. It combines a solidly negative correlation (-0.39 full, -0.56 in the drawdown) with the strongest absolute performance of any sector measured: +11.9% over the window and +8.9% during the chip decline. Financials deserve attention for a different reason — the lowest volatility of any sector here at 13.9%, a negative correlation that strengthens in the drawdown (-0.35), and +11.3% over the window, helped by record big-bank trading and dealmaking revenue.

Energy is the outlier that needs a warning label. Its -0.29 correlation is real and its +8.0% drawdown return is the second best measured, but the mechanism is not defensiveness — it is a geopolitical supply shock, with Brent pushed toward $100 by Houthi tanker attacks and Iran escalation, and record US refining margins. That is an independent factor, which diversifies chips, but it is a live directional bet on oil rather than a shock absorber. It is the AEM/WPM lesson from the previous screen in a new costume: uncorrelated to chips, fully correlated to something else.

5 THE PATTERNS — WHAT THE SECTOR RESULTS ACTUALLY SAY

ELI5 — Why the old 'defensive' label has stopped working

For decades, if you wanted something that would not fall with technology, you bought utilities — dull, regulated, bond-like. But AI data centres need enormous amounts of electricity, so the market has quietly reclassified power companies as part of the AI trade. The same has happened to industrial companies that make turbines, cooling, switchgear and electrical equipment. They still LOOK defensive on the label, but their share prices now follow the AI capital-spending story. Buying them as a chip hedge is like buying a fire extinguisher made of firewood.

PATTERN ONE — the hedge is not 'defensive sectors', it is 'sectors with no AI capex linkage'. This is the most useful single finding in the study, and it is what the top-down approach exposed that a name-by-name screen would have missed. Industrials measures +0.62 to SOXX, essentially trading as an AI-infrastructure proxy, and utilities has been dragged from classic defensive to a weak -0.15 by the same force. Both would have been picked by a conventional defensive screen; both would have failed as chip hedges. What survives — staples, health care, insurance-type financials — are the sectors whose revenue has no line item that gets bigger when a data centre is built.

PATTERN TWO — the demand must be non-deferrable and the pricing non-cyclical. Every sector that hedged well sells something a customer cannot postpone: food, household products, prescriptions, insurance cover. That produces revenue which neither rises nor falls with the AI capital cycle, so when capital rotates out of chips these cash flows become the destination rather than the source. Consumer discretionary is the control case that proves it — nominally a consumer sector, but measuring +0.34 because its index weight is dominated by a handful of mega-cap technology-adjacent names, and it posted the worst return in the study at -6.4%.

PATTERN THREE — the correlations strengthen when it matters. In four of the five negatively-correlated sectors the correlation became MORE negative during the drawdown than over the full window (staples -0.63 to -0.77, health care -0.39 to -0.56, financials -0.26 to -0.35, utilities -0.15 to -0.33). This asymmetry is the property worth paying for: these are not merely unrelated to chips, they are the specific destination for capital leaving chips. Industrials, by contrast, held its positive correlation right through the decline (+0.62 to +0.64), confirming it is inside the trade rather than opposite it.

6 NAME LAYER — RANKED BY CORRELATION TO SOXX

Only now do we descend to individual names. These correlations to SOXX come from the 13-name universe measured in the previous screen over a slightly earlier window (29 April to 13 July), which includes the first two weeks of the drawdown but not the latest leg; the window difference is flagged in the Verification Log and the ranking should be read as indicative rather than precise. The 'Sector read' column connects each name back to the top-down finding.

Ticker	Name	Corr SOXX	Sector	Why it behaves that way
PGR	Progressive	-0.66	Financials	Insurance underwriting cycle; no AI linkage
MCK	McKesson	-0.65	Health care	Fee-per-script distribution oligopoly
WRB	W. R. Berkley	-0.42	Financials	Specialty insurance — FLAGGED, thin data
PM	Philip Morris	-0.38	Staples	Price-inelastic global nicotine cash flows
CME	CME Group	-0.38	Financials	Exchange fees; volatility is its product
NOC	Northrop Grumman	-0.33	Industrials	Defense — the non-AI part of a +0.62 sector
ORLY	O'Reilly Auto	-0.28	Discretionary	Aftermarket — non-AI part of a +0.34 sector
BRK.B	Berkshire Hathaway	-0.27	Financials	Diversified conglomerate; lowest volatility
WMT	Walmart	-0.21	Staples	Non-deferrable consumer demand
GILD	Gilead Sciences	-0.10	Health care	HIV franchise annuity; near-zero, not hedge
TPL	Texas Pacific Land	+0.23	Energy	Permian royalty — energy factor, not a hedge
AEM	Agnico Eagle	+0.48	Materials	Gold-factor bet; fell WITH risk
WPM	Wheaton Precious	+0.61	Materials	Gold-factor bet; worst crash-day record

The two strongest names, Progressive at -0.66 and McKesson at -0.65, are more negatively correlated to chips than any sector index — which is what you would expect, since a sector ETF dilutes the pure mechanism across dozens of holdings. Progressive's earnings depend on premium rates and claims frequency; McKesson earns a contracted margin on each prescription distributed. Neither has any revenue line that responds to data-centre construction, which is precisely the Pattern One criterion.

The most useful subtlety sits in the last two sector rows of the table. Industrials as a whole is +0.62, but Northrop Grumman inside it measures -0.33, because defense procurement runs on multi-year government budgets rather than AI capex. Consumer discretionary is +0.34, but O'Reilly inside it measures -0.28, because the auto aftermarket is counter-cyclical. The lesson is that the AI-linkage test operates at SUB-INDUSTRY level: a positively-correlated sector can still contain genuine hedges, and a defensive-labelled sector can contain AI proxies. Sector ETFs are a good map and a poor instrument.

Names to avoid for this purpose are equally clear. The gold miners at +0.48 and +0.61 are not hedges to anything — they are a leveraged position in the gold price, and they fell hardest of anything measured on the worst chip days. Texas Pacific Land at +0.23 carries the energy factor rather than a defensive one. Gilead at -0.10 is genuinely uncorrelated rather than negatively correlated: excellent as a diversifier, but it will not actively rise to offset a chip drawdown.

7 CONSTRUCTION — THE SIZING PROBLEM NOBODY MENTIONS

ELI5 — Why a dollar of hedge does not offset a dollar of chips

Imagine trying to balance a see-saw with a jumping child on one end and a sleeping cat on the other. Even if the cat reliably moves the opposite way, it barely moves at all — so it takes a great many cats to balance one child. Semiconductors move roughly three to four times as violently as staples or health care. Correlation tells you the cat is on the right side of the see-saw; it says nothing about how many cats you need.

The volatility asymmetry is severe and is the practical constraint on this entire exercise. SOXX ran at 67.0% annualised volatility over the window against 17.8% for staples, 18.8% for health care and 13.9% for financials — a ratio of roughly 3.6 to 4.8 times. This is why the measured betas to SOXX look so small (-0.16 for staples, -0.11 for health care): a strongly negative CORRELATION combined with far lower volatility produces only a modest offset per dollar. To neutralise a meaningful share of chip drawdown risk with staples would require a hedge notional several times the size a naive reading of the correlation would suggest.

That argues for treating this as a portfolio tilt rather than an engineered hedge. A blend across staples, health care and insurance-type financials both diversifies the idiosyncratic risk within the sleeve and captures the property that matters — all three became more negatively correlated during the drawdown. Health care is the natural core given it delivered the best absolute return of any sector while doing so; staples is the purest hedge by correlation; financials adds the lowest volatility, which improves the sleeve's own stability.

Two construction notes. First, sizing should reflect that these positions are meant to be held through the regime, not traded around it: the negative correlation is a product of rotation flows, and attempting to time entry into a flow that is already running risks buying the destination after the money has arrived — health care is already up 11.9%. Second, energy should be sized as a separate directional oil position rather than counted as part of the defensive sleeve, because its diversification is a coincidence of the current geopolitical shock rather than a structural property.

8 RISKS — WHAT WOULD BREAK THIS

The rotation becomes a crash. Everything measured here is capital moving between equity sectors while the aggregate market holds. If the chip unwind broadens into a genuine growth scare or a funding event, correlations across equities converge toward positive and the staples and health care positions fall alongside the chips, simply less. The specific trigger to watch is the AI capex narrative shifting from 'pause' to 'over': that would take the industrial and power complex down with semiconductors and remove the destination that is currently bidding for defensive cash flows. This screen's hedges are rotation hedges, and only rotation hedges.

The rotation reverses and you are on the wrong side. The same mechanism that makes staples negatively correlated on the way down makes it negatively correlated on the way back up. Semiconductor ETFs remain up very substantially year-to-date even after a 19.5% decline, and both Barclays and other desks have stayed constructive on the group while others have not; the Kospi rebounded 6% in a single session late in the window and SOXX itself rallied 5.45% on 21 July. If chips resume leadership, this sleeve becomes the funding source and underperforms precisely when the main book recovers. That is the intended trade-off, but it should be an explicit decision rather than a surprise.

Utilities and industrials could re-decouple, or decouple further. The -0.15 utilities reading and +0.62 industrials reading are recent phenomena driven by the AI power and equipment narrative. If that narrative breaks, these sectors may revert toward their historical defensive behaviour and become useful hedges again — or, if AI capex accelerates, industrials could become an even purer chip proxy. Either way the classification in this report is a snapshot of a moving relationship and should be re-measured, not assumed to hold for quarters.

Sample and window limitations. The sector figures rest on 48 daily observations, with only 24 in the drawdown sub-period, giving standard errors of roughly 0.14 and 0.20 respectively — the ordering of adjacent sectors such as energy at -0.29 and financials at -0.26 is not meaningful, only the broad grouping is. The name-level correlations come from a slightly earlier window and are indicative. Four sectors were not measured at all (materials, real estate, technology, communication services); technology would obviously be strongly positive, and real estate and materials remain genuinely untested here.

Idiosyncratic risk inside the hedge. Buying a hedge does not exempt you from company problems: CME faces competitive pressure from Kalshi and perpetual-futures entrants, O'Reilly carries deal risk from its reported bid for Genuine Parts' NAPA unit, Philip Morris took a Canada impairment that cut guidance, and health care as a sector carries drug-pricing and policy exposure that has nothing to do with semiconductors. A hedge that works on correlation and fails on fundamentals still loses money.

9 VALUATION CONTEXT — WHAT THE HEDGE COSTS

A hedge that has already run is a more expensive hedge. Health care has gained 11.9% and financials 11.3% over the measurement window, so a portion of the rotation is in the price; staples at +1.1% and energy at +1.2% have moved far less on a full-window basis despite their strong drawdown performance, because both were being sold during the melt-up phase. Market capitalisations are ranges, and multiples are approximate vendor figures flagged EST. in the Verification Log.

Instrument	Mkt cap	Corr SOXX	Window ret % / P-E	Note
XLP staples	n/a (ETF)	-0.63	+1.1	Cheapest entry of the hedges; least run
XLV health care	n/a (ETF)	-0.39	+11.9	Best performer; rotation partly priced in
XLF financials	n/a (ETF)	-0.26	+11.3	Lowest volatility; bank earnings strong
PGR Progressive	$130-140bn	-0.66	~12x	Cheapest multiple among the named hedges
MCK McKesson	$100-115bn	-0.65	~22x	Distribution oligopoly; steady compounder
PM Philip Morris	$275-285bn	-0.38	~25x	High yield; total return understated by price
BRK.B Berkshire	~$1.07tn	-0.27	n/m	Ballast; 14.1% vol, lowest in prior study
GILD Gilead	$155-185bn	-0.10	~19x	Diversifier not hedge; near-zero correlation

Bull case for putting the sleeve on now: the drawdown is chip-specific, the rotation destination is visible in the data rather than hypothesised, four of five defensive sectors became MORE negatively correlated as the decline deepened, and staples in particular has barely participated in the rotation on a full-window basis, so the cheapest hedge is also the most negatively correlated one. Progressive at roughly 12 times earnings offers the mechanism at the lowest multiple of any name measured.

Bear case: you are buying insurance after the fire has started. Health care and financials have already delivered double-digit gains in ten weeks, chips remain up enormously year-to-date and a 19.5% decline in a group that rose over 80% in a half-year is an unwind rather than a break, and the entire negative-correlation property depends on a rotation regime that could reverse in a single session — as it nearly did on 21 July when SOXX rose 5.45%. Buying the destination of a flow that has already arrived is the classic way to be right about the mechanism and wrong about the money.

A APPENDIX — VERIFICATION LOG

Item	Status	Basis
Sector ETF daily series (7 sectors + SOXX)	CLEAN	stockanalysis.com history pages, S&P Global Market Intelligence; adjusted close; all 8 spot-checks of computed daily returns vs published change matched exactly
Sector correlations, betas, volatility, drawdown sub-period	CLEAN	Computed bottom-up in pandas from container CSVs; n=48 full window, n=24 drawdown
SOXX drawdown magnitude (-19.5% from 22 Jun peak)	CLEAN	Computed from the fetched series; peak 655.01 on 22 Jun to 527.01 on 24 Jul
Name-level correlations to SOXX	EST.	Carried from the prior screen's window (29 Apr-13 Jul); does not include the final drawdown leg — window mismatch flagged
WRB correlation (-0.42)	FLAGGED	Only 9 overlapping observations in the source window; indicative only, not ranked
Regime diagnosis (rotation vs systemic)	EST.	Multi-source financial press plus our own measured sector returns; the measured returns are CLEAN, the narrative attribution is interpretive
Drawdown catalysts (Broadcom, SK Hynix, Intel 18A, Meta compute)	EST.	Financial press reports; not independently verified against filings
Market caps and P/E ratios	EST.	Ranges across aggregators; vendor sampling dates vary
Unmeasured sectors (XLB, XLRE, XLK, XLC)	FLAGGED	Not fetched; excluded from all conclusions rather than estimated

Method caveats: adjusted close is split- and dividend-adjusted, so figures approximate total return; sector ETF windows end 27 July while SOXX's vendor cache ends 24 July, handled by pairwise overlap; correlation standard error is roughly 0.14 on the full window and 0.20 on the drawdown sub-period; single-vendor concentration for price data (S&P Global Market Intelligence via stockanalysis.com) is acknowledged. All figures describe one regime and are not forecasts.

B APPENDIX — ACRONYM GLOSSARY

Acronym	Full name / meaning
SOXX	iShares Semiconductor ETF — the chip complex benchmark used throughout
ETF	Exchange-Traded Fund — a listed fund tracking an index or sector
SPDR	Standard & Poor's Depositary Receipts — State Street's ETF family
XLP	Consumer Staples Select Sector SPDR ETF
XLV	Health Care Select Sector SPDR ETF
XLE	Energy Select Sector SPDR ETF
XLF	Financial Select Sector SPDR ETF
XLU	Utilities Select Sector SPDR ETF
XLI	Industrial Select Sector SPDR ETF
XLY	Consumer Discretionary Select Sector SPDR ETF
XLB / XLRE / XLK / XLC	Materials / Real Estate / Technology / Communication Services SPDRs — not measured here
SOX	Philadelphia Semiconductor Index — the underlying chip index
S&P 500	Standard & Poor's 500 Index
AI	Artificial Intelligence

Acronym	Full name / meaning
HBM	High-Bandwidth Memory — stacked DRAM used in AI accelerators
DRAM	Dynamic Random-Access Memory
Capex	Capital Expenditure — spending on physical assets such as data centres
18A	Intel's 1.8-nanometre-class foundry process node
NAPA	National Automotive Parts Association — Genuine Parts' auto-parts arm
P/E	Price-to-Earnings ratio
YTD	Year-To-Date
EST.	Estimate — from secondary sources, not independently re-computed
n	Number of overlapping daily observations in a pairwise correlation

C APPENDIX — DISCLAIMER